

DESIGN DOCUMENT

Course : Management of AI Products

Assignment: Experimental Learning Capstone

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Title: AI- Assisted Ticket Resolution System

1. Project Overview

Support teams often spend significant time searching through product documentation and previously resolved client tickets to find solutions for new issues. Even when a similar problem has already been solved, locating the right information can be difficult and time-consuming.

This project proposes an AI-based Support Assistant that can search existing documentation and historical support tickets and suggest possible solutions for a new client issue.

2. How It Works

The system will use a **Retrieval-Augmented Generation (RAG)** approach:

Knowledge Sources:

- Product and technical documentation
- FAQs and troubleshooting guides
- Previously resolved client tickets
- Known issues and their solutions

These sources will be divided into smaller sections and converted into embeddings, which will be stored in a vector database.

Issue Retrieval:

When a new client ticket is submitted, the issue will also be converted into an embedding. Vector similarity search / K-Nearest Neighbours (KNN) will then be used to find the most relevant documentation and similar historical tickets.

The retrieved information will be passed to the language model as context. The model will use this information to generate a suggested solution rather than relying only on its general knowledge.

3. Handling Missing Information

The assistant should not provide a solution when the available information is insufficient.

Instead, it should identify what information is missing and ask the user relevant follow-up questions, such as:

- What error message are you receiving?
- Which application or API version are you using?
- Which API endpoint or feature is affected?
- What steps were performed before the issue occurred?

After receiving the additional information, the system can perform the search again and improve its suggestion.

4. Proposed System Flow

Client Issue → Query/Embedding → KNN/Vector Search → Relevant Documentation & Previous Tickets → RAG + LLM → Suggested Solution / Follow-up Questions

The response can also include the documentation sections or previous ticket references used to generate the suggestion, allowing the support engineer to verify the recommendation.

5. Expected Outcome

The goal is to reduce the time spent investigating support issues and make knowledge from previously resolved tickets easier to reuse. The assistant will act as a support tool rather than replacing the support engineer.

The engineer will remain responsible for reviewing the suggested solution and deciding whether it should be provided to the client.

6. Key Technologies

RAG • Embeddings • Vector Database • KNN/Similarity Search • Large Language Model (LLM) • Document & Ticket Processing